

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2_AT2 — VISUALIZATION DEVELOPMENT EXERCISE

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Development (BL4)	Assessment:	Assessment Tool 2 – Visualization Development
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO2 Statement:	<i>Analyze and synthesize multiple data views into a coherent, annotated dashboard that communicates a clear narrative insight. (BL4)</i>		
Student Name:	M.SAI HARSHITHA	Reg. No.:	192424408
Section / Batch:	B.TECH AI & DS	Date:	30.8.2026

Code of Conduct

I M.SAI HARSHITHA/192424408(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.SAI HARSHITHA

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must include at least two coordinated charts sharing a common axis or theme.
- Include at least one annotation calling out a key data point in your dashboard.
- Write a one-sentence narrative takeaway summarizing the insight your dashboard reveals.

Annexure A – VISUALIZATION DEVELOPMENT

Rubrics for Calculation ASSESSMENT RUBRIC
AT2 — Visualization Development Exercise

CO2 · BT4: Analyze ·

This rubric is used to assess student responses to the CO2-AT2 Visualization Development Exercise, in which students build a small multi-chart dashboard from given data, add at least one annotation, and write a narrative takeaway.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of which chart types to combine, how they relate to each other, and how annotation and narrative should be used to communicate insight.	Good understanding of chart combination and annotation/narrative technique, with only minor gaps.	Basic understanding of dashboard development, but with some misconceptions about which charts to combine or how to annotate them.	Poor understanding of core development concepts; combines charts or annotations that do not suit the data or the story being told.
Explanation	25%	The annotation and narrative takeaway are precise, well-justified, and clearly tied to a specific, correctly identified data point or pattern.	The annotation and narrative are clear and reasonably well-justified, with adequate connection to the data.	The annotation or narrative is present but vague, generic, or weakly connected to the actual data shown.	Annotation or narrative is missing, incorrect, or unconnected to the data in the dashboard.
Application	20%	Effectively applies multi-chart coordination, annotation, and storytelling	Applies development techniques to produce a workable dashboard, with only minor errors or	Limited application; the dashboard partially addresses the scenario but charts feel	Fails to apply appropriate development techniques; the charts, annotation, and
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		techniques to produce a complete, cohesive dashboard for the scenario.	missed refinements.	disconnected or the story is unclear.	narrative do not work together as a dashboard.

Organization	15%	Charts are clearly coordinated (aligned axes or shared theme), laid out logically, with the annotation and narrative placed for maximum clarity.	Charts are organized and mostly coordinated, with only minor issues in alignment or layout.	Basic structure; charts are present but poorly aligned or the annotation/narrative feels disconnected from the layout.	Poorly organized; charts are not coordinated, and the annotation or narrative is missing or misplaced.
Accuracy	10%	All chart values, scales, and the annotated data point are completely accurate and correctly reflect the given data.	Mostly accurate, with only minor omissions or a small error in one chart or the annotation.	Partially correct; some plotted values or the annotated point are missing or incorrect.	Largely inaccurate; major errors in plotted values, scales, or the annotated data point.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

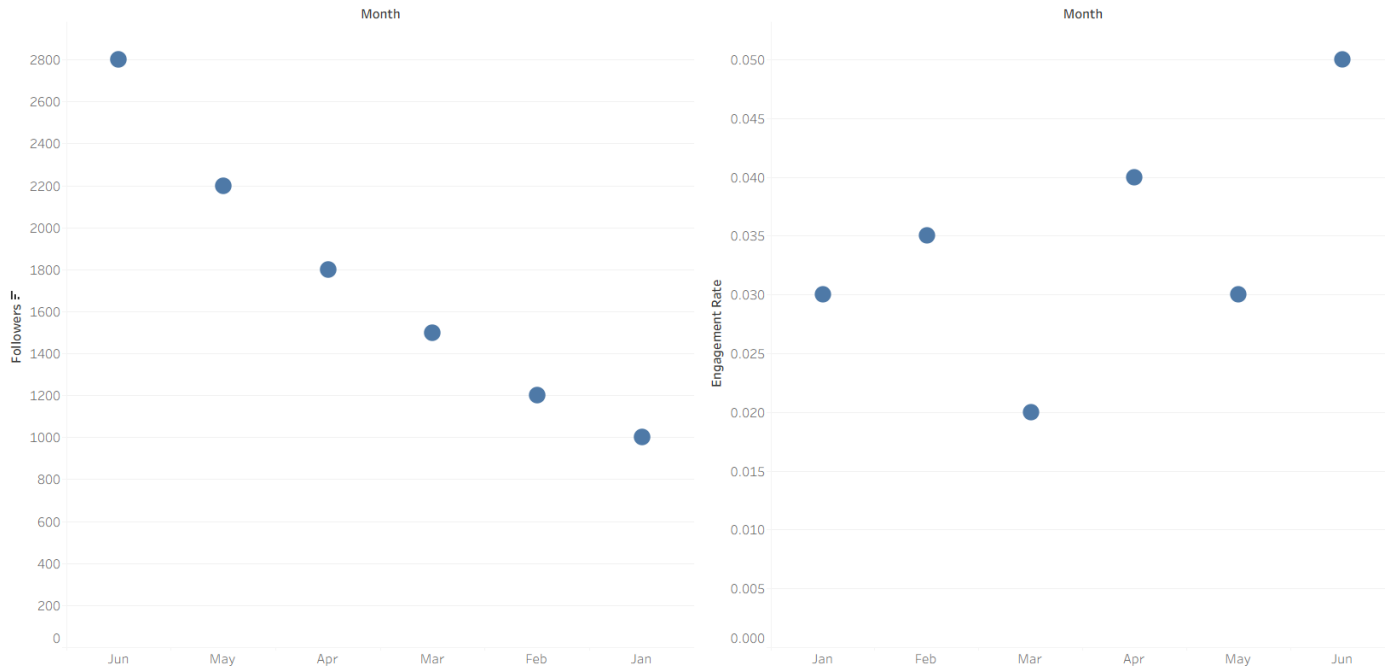
Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 6 — Social Media Growth Dashboard (5 Marks)

Data: Monthly followers and engagement rate: Jan = 1,000 / 3%, Feb = 1,200 / 3.5%, Mar = 1,500 / 2%, Apr = 1,800 / 4%, May = 2,200 / 3%, Jun = 2,800 / 5%.

Task: Build a dashboard with a line chart of follower growth and a line chart of engagement rate on the same month axis (stacked, not dual-axis). Annotate the month where engagement diverged notably from follower growth, and write a one-sentence narrative interpreting this.

March — Engagement rate dropped to 2% while followers increased to 1,500.



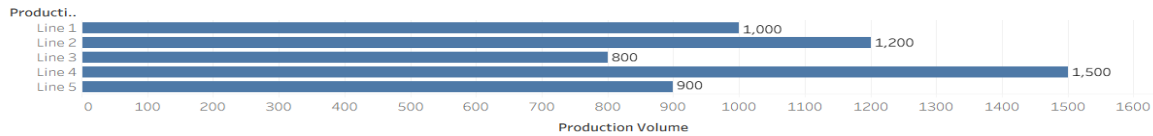
Follower growth continued, but engagement decreased significantly in March.

Problem 7 — Manufacturing Quality Dashboard (5 Marks)

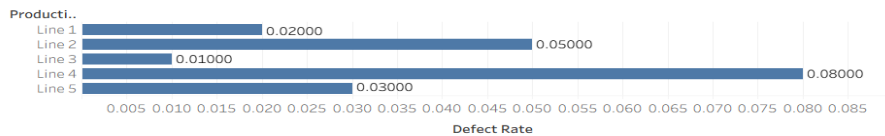
Data: Defect rate and production volume by line: Line 1 = 2% / 1,000 units, Line 2 = 5% / 1,200, Line 3 = 1% / 800, Line 4 = 8% / 1,500, Line 5 = 3% / 900.

Task: Build a dashboard with a bar chart of production volume and a bar chart of defect rate for the same lines. Annotate the line requiring immediate quality review, and add a takeaway for the plant manager.

Line 4 — Highest defect rate (8%).



Sheet 2



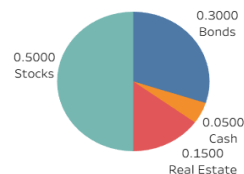
Line 4 requires immediate quality inspection due to its high defect rate.

Problem 8 — Financial Portfolio Dashboard (5 Marks)

Data: Allocation and annual return by asset class: Stocks = 50% / 12% return, Bonds = 30% / 4%, Real Estate = 15% / 8%, Cash = 5% / 1%.

Task: Build a dashboard combining a pie chart of allocation and a bar chart of returns for the same asset classes. Annotate the class with the best risk-adjusted contribution, and write a narrative recommending a rebalancing action.

Stocks — Highest contribution with 50% allocation and 12% return.



Sheet 2

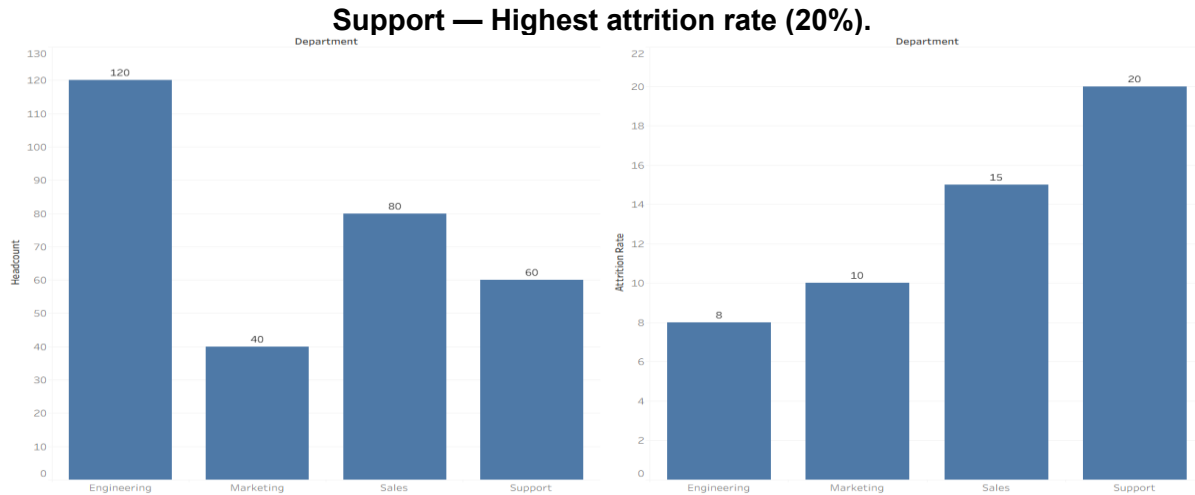


Increase diversification by balancing high-growth stocks with stable assets

Problem 9 — HR Attrition Dashboard (5 Marks)

Data: Headcount and attrition rate by department: Engineering = 120 / 8%, Sales = 80 / 15%, Support = 60 / 20%, Marketing = 40 / 10%.

Task: Build a dashboard with a bar chart of headcount and a bar chart of attrition rate for the same departments. Annotate the department needing urgent retention action, and write a one-sentence executive takeaway.

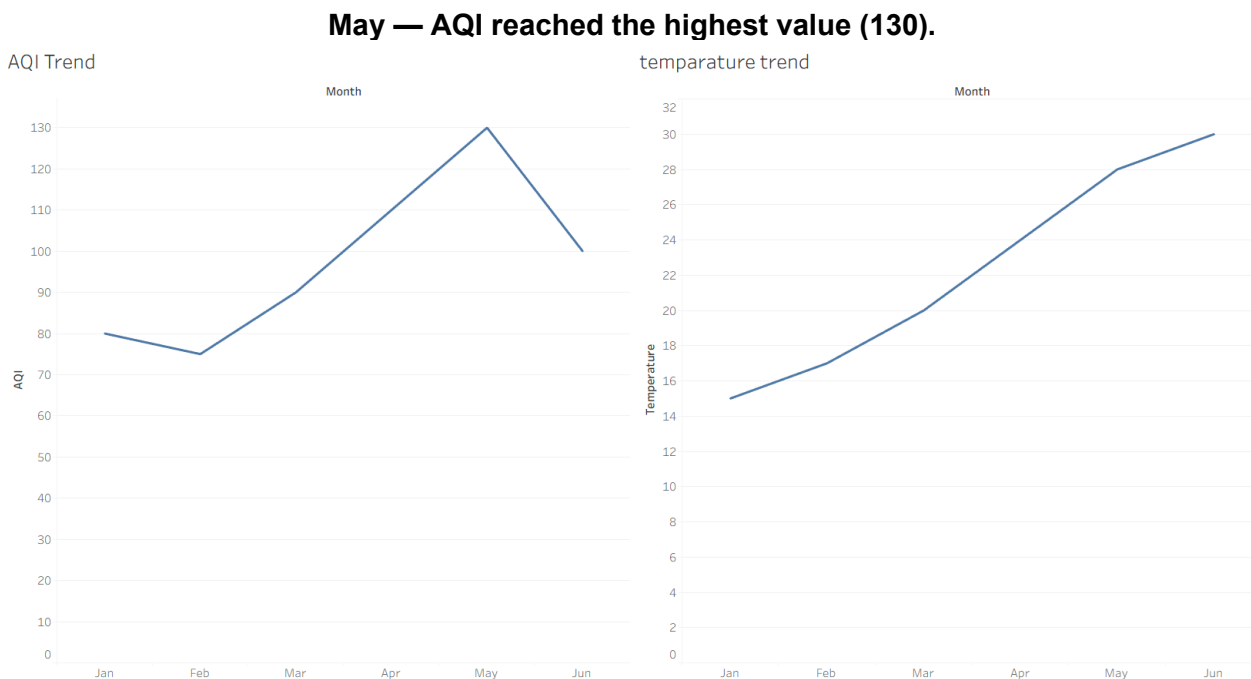


Support department needs immediate retention strategies due to high employee turnover

Problem 10 — Environmental Monitoring Dashboard (5 Marks)

Data: Monthly air quality index (AQI) and average temperature: Jan = 80 / 15°C, Feb = 75 / 17, Mar = 90 / 20, Apr = 110 / 24, May = 130 / 28, Jun = 100 / 30.

Task: Build a dashboard with a line chart of AQI and a line chart of temperature on the same month axis (stacked, not dual-axis). Annotate the month where AQI peaked, and write a narrative caption connecting temperature and air quality trends.



AQI increased with rising temperature and peaked in May.